Topic: Convolution integrals for initial value problems

Question: Use a convolution integral to find the general solution $y(t)$ to the differential equation, given $y(0) = 1$ and $y'(0) = 1$.

$$y'' + 2y = g(t)$$

Answer choices:

A $$y(t) = \frac{\sqrt{2}}{2}\left(\sqrt{2}\cos(\sqrt{2}t) + \sin(\sqrt{2}t) + \int_0^t \sin(\sqrt{2}\tau)g(t+\tau)\ d\tau\right)$$

B $$y(t) = \frac{\sqrt{2}}{2}\left(\sqrt{2}\cos(\sqrt{2}t) + \sin(\sqrt{2}t) + \int_0^t \sin(\sqrt{2}\tau)g(t-\tau)\ d\tau\right)$$

C $$y(t) = \frac{\sqrt{2}}{2}\left(\sqrt{2}\cos(\sqrt{2}t) + \sin(\sqrt{2}t) + \int_0^t \cos(\sqrt{2}\tau)g(t+\tau)\ d\tau\right)$$

D $$y(t) = \frac{\sqrt{2}}{2}\left(\sqrt{2}\cos(\sqrt{2}t) + \sin(\sqrt{2}t) + \int_0^t \cos(\sqrt{2}\tau)g(t-\tau)\ d\tau\right)$$

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Solution: B

From a table of Laplace transforms, we know that

$$\mathscr{L}(y'') = s^2Y(s) - sy(0) - y'(0)$$

$$\mathscr{L}(y) = Y(s)$$

$$\mathscr{L}(g(t)) = G(s)$$

Making substitutions into the differential equation gives

$$s^2Y(s) - sy(0) - y'(0) + 2Y(s) = G(s)$$

Now we'll plug in the initial conditions $y(0) = 1$ and $y'(0) = 1$ in order to simplify the transform.

$$s^2Y(s) - s(1) - 1 + 2Y(s) = G(s)$$

$$s^2Y(s) - s - 1 + 2Y(s) = G(s)$$

We'll solve for $Y(s)$ by gathering all the $Y(s)$ terms on the left, and moving all other terms to the right, then factoring out a $Y(s)$.

$$s^2Y(s) + 2Y(s) = G(s) + s + 1$$

$$Y(s)(s^2 + 2) = G(s) + s + 1$$

$$Y(s) = \frac{s}{s^2 + 2} + \frac{1}{s^2 + 2} + G(s)\left(\frac{1}{s^2 + 2}\right)$$

We want to use an inverse Laplace transform to put each part of this equation in terms of t instead of s. If we start with the first term, we can see its similarity to the formula for the transform of $\cos(at)$.

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$$\mathscr{L}(\cos(at)) = \frac{s}{s^2 + a^2}$$

If we say $a = \sqrt{2}$, then the inverse transform of that first term is

$$\mathscr{L}^{-1}\left(\frac{s}{s^2 + 2}\right) = \cos(\sqrt{2}t)$$

The second term from $Y(s)$ should remind us of the formula for the transform of $\sin(at)$.

$$\mathscr{L}(\sin(at)) = \frac{a}{s^2 + a^2}$$

Let's rewrite the second term so that it better matches the transform formula for $\sin(at)$.

$$\frac{1}{s^2 + 2}$$

$$\frac{\frac{\sqrt{2}}{\sqrt{2}}}{s^2 + (\sqrt{2})^2}$$

$$\frac{\frac{1}{\sqrt{2}}\sqrt{2}}{s^2 + (\sqrt{2})^2}$$

$$\frac{1}{\sqrt{2}}\left(\frac{\sqrt{2}}{s^2 + (\sqrt{2})^2}\right)$$

Now with $a = \sqrt{2}$, the inverse transform of that second term is

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$$\mathscr{L}^{-1}\left(\frac{1}{s^2+2}\right)=\frac{1}{\sqrt{2}}\sin(\sqrt{2}t)$$

Finding the inverse transform of the last term needs the convolution integral. We already know

$$\mathscr{L}^{-1}\left(\frac{1}{s^2+2}\right)=\frac{1}{\sqrt{2}}\sin(\sqrt{2}t)$$

and we can say that the inverse transform of $G(s)$ is $g(t)$, so for our convolution integral, we'll use the functions

$$f(t)=\frac{1}{\sqrt{2}}\sin(\sqrt{2}t)$$

$$g(t)=g(t)$$

Plugging these into the convolution integral, we get

$$f(t)*g(t)=\int_0^t f(\tau)g(t-\tau)\,d\tau$$

$$f(t)*g(t)=\frac{1}{\sqrt{2}}\int_0^t \sin(\sqrt{2}\tau)g(t-\tau)\,d\tau$$

Plugging all of these values back into the equation for $Y(s)$,

$$Y(s)=\frac{s}{s^2+2}+\frac{1}{s^2+2}+G(s)\left(\frac{1}{s^2+2}\right)$$

gives us the inverse transform, which is the general solution to the second order differential equation initial value problem.

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$$y(t) = \cos(\sqrt{2}t) + \frac{1}{\sqrt{2}}\sin(\sqrt{2}t) + \frac{1}{\sqrt{2}}\int_0^t \sin(\sqrt{2}\tau)g(t-\tau)\ d\tau$$

$$y(t) = \frac{1}{\sqrt{2}}\left(\sqrt{2}\cos(\sqrt{2}t) + \sin(\sqrt{2}t) + \int_0^t \sin(\sqrt{2}\tau)g(t-\tau)\ d\tau\right)$$

$$y(t) = \frac{\sqrt{2}}{2}\left(\sqrt{2}\cos(\sqrt{2}t) + \sin(\sqrt{2}t) + \int_0^t \sin(\sqrt{2}\tau)g(t-\tau)\ d\tau\right)$$

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Topic: Convolution integrals for initial value problems

Question: Use a convolution integral to find the general solution $y(t)$ to the differential equation, given $y(0) = 1$ and $y'(0) = 1$.

$$3y'' + y = g(t)$$

Answer choices:

A $$y(t) = \cos\left(\frac{\sqrt{3}}{3}t\right) + \sqrt{3}\sin\left(\frac{\sqrt{3}}{3}t\right) + \int_0^t \cos\left(\frac{\sqrt{3}}{3}\tau\right) g(t+\tau)\, d\tau$$

B $$y(t) = \cos\left(\frac{\sqrt{3}}{3}t\right) + \sqrt{3}\sin\left(\frac{\sqrt{3}}{3}t\right) + \int_0^t \sin\left(\frac{\sqrt{3}}{3}\tau\right) g(t-\tau)\, d\tau$$

C $$y(t) = \cos\left(\frac{\sqrt{3}}{3}t\right) + \sqrt{3}\sin\left(\frac{\sqrt{3}}{3}t\right) + \sqrt{3}\int_0^t \sin\left(\frac{\sqrt{3}}{3}\tau\right) g(t+\tau)\, d\tau$$

D $$y(t) = \cos\left(\frac{\sqrt{3}}{3}t\right) + \sqrt{3}\sin\left(\frac{\sqrt{3}}{3}t\right) + \frac{\sqrt{3}}{3}\int_0^t \sin\left(\frac{\sqrt{3}}{3}\tau\right) g(t-\tau)\, d\tau$$

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Solution: D

From a table of Laplace transforms, we know that

$$\mathscr{L}(y'') = s^2Y(s) - sy(0) - y'(0)$$

$$\mathscr{L}(y) = Y(s)$$

$$\mathscr{L}(g(t)) = G(s)$$

Making substitutions into the differential equation gives

$$3[s^2Y(s) - sy(0) - y'(0)] + Y(s) = G(s)$$

Now we'll plug in the initial conditions $y(0) = 1$ and $y'(0) = 1$ in order to simplify the transform.

$$3[s^2Y(s) - s(1) - 1] + Y(s) = G(s)$$

$$3s^2Y(s) - 3s - 3 + Y(s) = G(s)$$

We'll solve for $Y(s)$ by gathering all the $Y(s)$ terms on the left, and moving all other terms to the right, then factoring out a $Y(s)$.

$$3s^2Y(s) + Y(s) = G(s) + 3s + 3$$

$$Y(s)(3s^2 + 1) = G(s) + 3s + 3$$

$$Y(s) = 3\left(\frac{s}{3s^2 + 1}\right) + 3\left(\frac{1}{3s^2 + 1}\right) + G(s)\left(\frac{1}{3s^2 + 1}\right)$$

$$Y(s) = 3\left(\frac{\frac{1}{3}s}{s^2 + \frac{1}{3}}\right) + 3\left(\frac{\frac{1}{3}}{s^2 + \frac{1}{3}}\right) + G(s)\left(\frac{\frac{1}{3}}{s^2 + \frac{1}{3}}\right)$$

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$$Y(s) = \frac{s}{s^2 + \frac{1}{3}} + \frac{1}{s^2 + \frac{1}{3}} + \frac{1}{3}G(s)\left(\frac{1}{s^2 + \frac{1}{3}}\right)$$

We want to use an inverse Laplace transform to put each part of this equation in terms of t instead of s. If we start with the first term, we can see its similarity to the formula for the transform of $\cos(at)$.

$$\mathscr{L}(\cos(at)) = \frac{s}{s^2 + a^2}$$

If we say $a = \sqrt{1/3} = 1/\sqrt{3} = \sqrt{3}/3$, then the inverse transform of that first term is

$$\mathscr{L}^{-1}\left(\frac{s}{s^2 + \frac{1}{3}}\right) = \cos\left(\frac{\sqrt{3}}{3}t\right)$$

The second term from $Y(s)$ should remind us of the formula for the transform of $\sin(at)$.

$$\mathscr{L}(\sin(at)) = \frac{a}{s^2 + a^2}$$

Let's rewrite the second term so that it better matches the transform formula for $\sin(at)$.

$$\frac{1}{s^2 + \frac{1}{3}}$$

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$$\frac{\frac{\frac{\sqrt{3}}{3}}{\frac{\sqrt{3}}{3}}}{s^2+\left(\frac{\sqrt{3}}{3}\right)^2}$$

$$\frac{\frac{1}{\frac{\sqrt{3}}{3}}\left(\frac{\sqrt{3}}{3}\right)}{s^2+\left(\frac{\sqrt{3}}{3}\right)^2}$$

$$\sqrt{3}\left(\frac{\frac{\sqrt{3}}{3}}{s^2+\left(\frac{\sqrt{3}}{3}\right)^2}\right)$$

Now with $a=\sqrt{3}/3$, the inverse transform of that second term is

$$\mathscr{L}^{-1}\left(\frac{1}{s^2+\frac{1}{3}}\right)=\sqrt{3}\sin\left(\frac{\sqrt{3}}{3}t\right)$$

Finding the inverse transform of the last term needs the convolution integral. We already know

$$\mathscr{L}^{-1}\left(\frac{1}{s^2+\frac{1}{3}}\right)=\sqrt{3}\sin\left(\frac{\sqrt{3}}{3}t\right)$$

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and we can say that the inverse transform of $G(s)$ is $g(t)$, so for our convolution integral, we'll use the functions

$$f(t) = \sqrt{3} \sin\left(\frac{\sqrt{3}}{3}t\right)$$

$$g(t) = g(t)$$

Plugging these into the convolution integral, we get

$$f(t) * g(t) = \int_0^t f(t)g(t-\tau)\, d\tau$$

$$f(t) * g(t) = \sqrt{3} \int_0^t \sin\left(\frac{\sqrt{3}}{3}\tau\right) g(t-\tau)\, d\tau$$

Plugging all of these values back into the equation for $Y(s)$,

$$Y(s) = \frac{s}{s^2 + \frac{1}{3}} + \frac{1}{s^2 + \frac{1}{3}} + \frac{1}{3}G(s)\left(\frac{1}{s^2 + \frac{1}{3}}\right)$$

gives us the inverse transform, which is the general solution to the second order differential equation initial value problem.

$$y(t) = \cos\left(\frac{\sqrt{3}}{3}t\right) + \sqrt{3} \sin\left(\frac{\sqrt{3}}{3}t\right) + \frac{\sqrt{3}}{3} \int_0^t \sin\left(\frac{\sqrt{3}}{3}\tau\right) g(t-\tau)\, d\tau$$

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Topic: Convolution integrals for initial value problems

Question: Use a convolution integral to find the general solution $y(t)$ to the differential equation, given $y(0) = 1$ and $y'(0) = -2$.

$$2y'' + 5y = g(t)$$

Answer choices:

A $$y(t) = \cos\left(\frac{\sqrt{10}}{2}t\right) - \frac{2\sqrt{10}}{5}\sin\left(\frac{\sqrt{10}}{2}t\right) + \frac{\sqrt{10}}{5}\int_0^t \sin\left(\frac{\sqrt{10}}{2}\tau\right)g(t-\tau)\,d\tau$$

B $$y(t) = \cos\left(\frac{\sqrt{10}}{2}t\right) - \frac{4\sqrt{10}}{5}\sin\left(\frac{\sqrt{10}}{2}t\right) + \frac{\sqrt{10}}{5}\int_0^t \sin\left(\frac{\sqrt{10}}{2}\tau\right)g(t-\tau)\,d\tau$$

C $$y(t) = \cos\left(\frac{\sqrt{10}}{2}t\right) - \frac{2\sqrt{10}}{5}\sin\left(\frac{\sqrt{10}}{2}t\right) + \frac{\sqrt{10}}{10}\int_0^t \sin\left(\frac{\sqrt{10}}{2}\tau\right)g(t-\tau)\,d\tau$$

D $$y(t) = \cos\left(\frac{\sqrt{10}}{2}t\right) - \frac{4\sqrt{10}}{5}\sin\left(\frac{\sqrt{10}}{2}t\right) + \frac{\sqrt{10}}{20}\int_0^t \sin\left(\frac{\sqrt{10}}{2}\tau\right)g(t-\tau)\,d\tau$$

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Solution: C

From a table of Laplace transforms, we know that

$$\mathscr{L}(y'') = s^2Y(s) - sy(0) - y'(0)$$

$$\mathscr{L}(y) = Y(s)$$

$$\mathscr{L}(g(t)) = G(s)$$

Making substitutions into the differential equation gives

$$2[s^2Y(s) - sy(0) - y'(0)] + 5Y(s) = G(s)$$

Now we'll plug in the initial conditions $y(0) = 1$ and $y'(0) = -2$ in order to simplify the transform.

$$2[s^2Y(s) - s(1) - (-2)] + 5Y(s) = G(s)$$

$$2s^2Y(s) - 2s + 4 + 5Y(s) = G(s)$$

We'll solve for $Y(s)$ by gathering all the $Y(s)$ terms on the left, and moving all other terms to the right, then factoring out a $Y(s)$.

$$2s^2Y(s) + 5Y(s) = G(s) + 2s - 4$$

$$Y(s)(2s^2 + 5) = G(s) + 2s - 4$$

$$Y(s) = 2\left(\frac{s}{2s^2 + 5}\right) - 4\left(\frac{1}{2s^2 + 5}\right) + G(s)\left(\frac{1}{2s^2 + 5}\right)$$

$$Y(s) = 2\left(\frac{\frac{1}{2}s}{s^2 + \frac{5}{2}}\right) - 4\left(\frac{\frac{1}{2}}{s^2 + \frac{5}{2}}\right) + G(s)\left(\frac{\frac{1}{2}}{s^2 + \frac{5}{2}}\right)$$

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$$Y(s) = \frac{s}{s^2 + \frac{5}{2}} - 2\left(\frac{1}{s^2 + \frac{5}{2}}\right) + \frac{1}{2}G(s)\left(\frac{1}{s^2 + \frac{5}{2}}\right)$$

We want to use an inverse Laplace transform to put each part of this equation in terms of t instead of s. If we start with the first term, we can see its similarity to the formula for the transform of $\cos(at)$.

$$\mathscr{L}(\cos(at)) = \frac{s}{s^2 + a^2}$$

If we say $a = \sqrt{5/2}$, then the inverse transform of that first term is

$$\mathscr{L}^{-1}\left(\frac{s}{s^2 + \frac{5}{2}}\right) = \cos\left(\sqrt{\frac{5}{2}}t\right)$$

The second term from $Y(s)$ should remind us of the formula for the transform of $\sin(at)$.

$$\mathscr{L}(\sin(at)) = \frac{a}{s^2 + a^2}$$

Let's rewrite the second term so that it better matches the transform formula for $\sin(at)$.

$$\frac{1}{s^2 + \frac{5}{2}}$$

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$$\frac{\frac{\sqrt{\frac{5}{2}}}{\sqrt{\frac{5}{2}}}}{s^2+\left(\sqrt{\frac{5}{2}}\right)^2}$$

$$\frac{\frac{1}{\sqrt{\frac{5}{2}}}\sqrt{\frac{5}{2}}}{s^2+\left(\sqrt{\frac{5}{2}}\right)^2}$$

$$\sqrt{\frac{2}{5}}\left(\frac{\sqrt{\frac{5}{2}}}{s^2+\left(\sqrt{\frac{5}{2}}\right)^2}\right)$$

Plug this rewritten term back into the equation for $Y(s)$.

$$Y(s)=\frac{s}{s^2+\frac{5}{2}}-2\cdot\sqrt{\frac{2}{5}}\left(\frac{\sqrt{\frac{5}{2}}}{s^2+\left(\sqrt{\frac{5}{2}}\right)^2}\right)+\frac{1}{2}G(s)\cdot\sqrt{\frac{2}{5}}\left(\frac{\sqrt{\frac{5}{2}}}{s^2+\left(\sqrt{\frac{5}{2}}\right)^2}\right)$$

$$Y(s)=\frac{s}{s^2+\frac{5}{2}}-\frac{2\sqrt{2}}{\sqrt{5}}\left(\frac{\sqrt{\frac{5}{2}}}{s^2+\left(\sqrt{\frac{5}{2}}\right)^2}\right)+\frac{\sqrt{2}}{2\sqrt{5}}G(s)\left(\frac{\sqrt{\frac{5}{2}}}{s^2+\left(\sqrt{\frac{5}{2}}\right)^2}\right)$$

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Now with $a = \sqrt{5/2}$, the inverse transform of everything inside the parentheses of the second term is

$$\mathscr{L}^{-1}\left\{\frac{\sqrt{\frac{5}{2}}}{s^2+\left(\sqrt{\frac{5}{2}}\right)^2}\right\} = \sin\left(\sqrt{\frac{5}{2}}t\right)$$

Finding the inverse transform of the last term needs the convolution integral. We already know

$$\mathscr{L}^{-1}\left\{\frac{\sqrt{\frac{5}{2}}}{s^2+\left(\sqrt{\frac{5}{2}}\right)^2}\right\} = \sin\left(\sqrt{\frac{5}{2}}t\right)$$

and we can say that the inverse transform of $G(s)$ is $g(t)$, so for our convolution integral, we'll use the functions

$$f(t) = \sin\left(\sqrt{\frac{5}{2}}t\right)$$

$$g(t) = g(t)$$

Plugging these into the convolution integral, we get

$$f(t) * g(t) = \int_0^t f(\tau)g(t-\tau)\,d\tau$$

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$$f(t) * g(t) = \int_0^t \sin\left(\sqrt{\frac{5}{2}}\tau\right) g(t-\tau)\, d\tau$$

Plugging all of these values back into the equation for $Y(s)$,

$$Y(s) = \frac{s}{s^2 + \frac{5}{2}} - \frac{2\sqrt{2}}{\sqrt{5}}\left(\frac{\sqrt{\frac{5}{2}}}{s^2 + \left(\sqrt{\frac{5}{2}}\right)^2}\right) + \frac{\sqrt{2}}{2\sqrt{5}} G(s)\left(\frac{\sqrt{\frac{5}{2}}}{s^2 + \left(\sqrt{\frac{5}{2}}\right)^2}\right)$$

gives us the inverse transform, which is the general solution to the second order differential equation initial value problem.

$$y(t) = \cos\left(\sqrt{\frac{5}{2}}t\right) - \frac{2\sqrt{2}}{\sqrt{5}} \sin\left(\sqrt{\frac{5}{2}}t\right) + \frac{\sqrt{2}}{2\sqrt{5}} \int_0^t \sin\left(\sqrt{\frac{5}{2}}\tau\right) g(t-\tau)\, d\tau$$

$$y(t) = \cos\left(\frac{\sqrt{10}}{2}t\right) - \frac{2\sqrt{10}}{5} \sin\left(\frac{\sqrt{10}}{2}t\right) + \frac{\sqrt{10}}{10} \int_0^t \sin\left(\frac{\sqrt{10}}{2}\tau\right) g(t-\tau)\, d\tau$$

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